

Jesse Clark

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Permanently eligible to work in Australia and in the United States.

Summary

Product-oriented, research-minded computer scientist.

Obsessed with cool technology like single-molecule sensors, social robots, and spaceflight.

- Developed product prototypes and demonstrations used to secure venture capital and grant funding.
- Produced visualizations and automated reports to help find insights in data for space mission teams, B2C sales departments, and scientific instruments.
- Improved productivity and reliability of data-driven research labs.
- Experience with regulatory review for spaceflight and medical devices.

Employment

Geneus Technologies, Mountain View, California

May 2023 – May 2024

Principal Algorithm Engineer

Established dry-lab operations at a new office for a Chinese biotechnology company (深圳今是科技有限公司).

- Hired a data science team, managed local projects, and reported to executives overseas.
- Developed primary analysis algorithms for a nanopore DNA sequencing instrument.
- Designed statistical models to infer hidden variables from raw time-series sensor data, and calculate Phred quality scores.
- Selected high-accuracy algorithms to produce neural network training data, and high-performance algorithms for use in production.

Python C++ PyTorch Machine Learning Data Science Signal Processing State Estimation Time Series Analysis DNA Sequencing

Roche Molecular Diagnostics, Santa Clara, California

December 2019 – May 2023

Principal Software Engineer

Developed software for [nanopore DNA sequencing](#) instruments. Adapted mature algorithms for the needs of new experiments, including molecule detection, level-calling, base-calling, quality metrics, and neural network training.

Coordinated data compatibility between wet-lab experimentalists, GPU-accelerated signal processing, machine learning model training, and statistical simulations. Created validation tools to enforce data compatibility agreements and prevent unnecessary processing or troubleshooting.

Python TypeScript C++ TensorFlow AirFlow Jenkins Data Science Machine Learning DNA Sequencing

University of Technology, Sydney, Ultimo, New South Wales

November 2014 – December 2019

Senior Software Engineer

Responsible for software operations for [The Innovation and Enterprise Research Lab](#).

- We won first place in the global [RoboCup@Home](#) competition. I was responsible for software system reliability, sequential task planning, and natural language processing.
- Implemented robot behaviors for experiments and demonstrations. Presented research results at conferences.
- Developed curriculum and delivered university lectures and masterclasses.

Python JavaScript Swift SVG iOS ROS Keras Machine Learning Human-Robot Interaction Teaching

Sydney Programming School, Mosman, New South Wales

April 2015 – December 2019

Head of Teaching

Delivered programming lessons at an after-school program, as it grew from zero to over 1,000 students.

Scratch JavaScript Arduino Python Processing Lego Mindstorms Robotics Cryptography Blockchain Game Design Teaching

Modus Ponens, LLC

Founded a new business to develop web and mobile applications for Silicon Valley startups. Led teams of local and remote designers and developers. Had contact with prominent venture capitalists.

- **Maverick Poker** 2014
- **OneLessDesk** 2014
- **Destination Software** 2013 – 2014
- **GoBoldly** 2013
- **Scanadu** 2012
- **SolarCity** 2012
- **Soundwave** 2011
- **Robot Panda Productions** 2010 – 2011
- **WoWGen** 2008

Node.JS JavaScript HTML REST Progressive Enhancement Riak Erlang iOS Objective-C C# Android Java Ruby on Rails
Parse.com Firebase.com Salesforce.com Distributed Systems GIS Augmented Reality

NASA Ames Research Center, Moffett Field, California

July 2007 – April 2011

Computer Scientist

Developed applications for space mission workflows for the Space Shuttle and International Space Station.

Perl MySQL Lucene HTML JavaScript LaTeX Human-Computer Interaction Contextual Inquiry Root Cause Analysis

NASA Goddard Space Flight Center, Greenbelt, Maryland

June 2004 – June 2007

Computer Engineer

Developed robotics and graphics software to simulate contact dynamics for the Hubble Space Telescope.

Recovered and visualized 1960s climate data from **Nimbus** and **TIROS** weather satellite archives.

C++ OpenGL Perl Mathematica FANUC Robotics GNC

Wheels of Zeus, Los Gatos, California

2003

Intern

PIC Assembly Electrical Engineering Linux Apache HTTPD

Institute for Computer Assisted Orthopaedic Surgery, Pittsburgh, Pennsylvania

May 2002 – July 2002

Intern

C++ Perl Tcl/Tk OpenGL Electrical Engineering Augmented Reality

Board Membership

Johnson and Louise H. Clark Charitable Foundation, Saratoga, California

April 2023 – Present

Treasurer

Managing financial accounts for a family-run charitable foundation.

Publications

See [curriculum vitae](#) for details.

GitHub link: <https://github.com/jjclark1982>

ResearchGate link: <https://www.researchgate.net/scientific-contributions/Jesse-Clark-2116266501>

Education

Carnegie Mellon University, Pittsburgh, Pennsylvania

May 2004

Bachelor of Science in Computer Science, Minor in Mathematical Sciences

(4-year degree with research component — equivalent to an Honours degree)

C++ Java Perl Mathematica Lisp OCaml x86 Assembly Prolog LaTeX Machine Learning Kernel programming TCP/IP
Digital Logic Cryptography Linear Algebra Cellular Automata Bioinformatics